

NEU C61: Brain, Mind, and Behavior

Fall 2026 | Tu/Fri 5 – 6:30pm | 100 Undergrad Academic Building

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Teaching Staff

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Course Information

Lectures	Tues/Fri 5 – 6:30p, 100 Undergrad Academic Bldg
Final Exam	Thursday December 17 th , 11:30am – 2:30pm
Course Website	tararaam.github.io/neu-c61
bCourses	https://bcourses.berkeley.edu/courses/1557311

Office Hours:

Staff	Date & Time	Location
Professor Raam	Mon. 9:30-10:30am	Zoom
Gretchen Huston, GSI	Wed. 1:10-2pm	Yali's Café (1920 Oxford St) or Zoom
Lauren Eckhardt, GSI	Mon. 1-2pm	TBD
Yuheng Feng, GSI	Wed. 10:15-11:15am	Zoom
Joseph Serafino, GSI	Thurs. 4:30-5:30pm	Zoom
Sam Reifenshtein, GSI	Tues. 11am-12pm	Yali's Café (1920 Oxford St)

Discussion Sections:

Section	Time	GSI	Location
101	Mon. 9am	Yuheng	107 UAB
102	Mon. 10am	Yuheng	209 UAB
103	Mon. 11am	Sam	221 UAB
104	Mon. 12pm	Joseph	105 Dwinelle
105	Mon. 1pm	Joseph	107 UAB
106	Mon. 2pm	Sam	105 Dwinelle
107	Mon. 3pm	Lauren	205 Dwinelle
108	Mon. 4pm	Joseph	205 Dwinelle
109	Wed. 9am	Yuheng	107 UAB
110	Wed. 3pm	Lauren	205 Dwinelle
111	Wed. 11am	Sam	229 UAB
112	Wed. 12pm	Gretchen	105 Dwinelle
113	Wed. 1pm	Gretchen	107 UAB
114	Wed. 2pm	Lauren	107 UAB

Important Dates:

Start of Instruction	Wednesday August 26 th
Last day to add or drop	Wednesday Sept 16 th
Last day to change grade options	Friday Oct 30 th
Last day of instruction	Friday Dec 11 th

Course Description

The human brain is the most complex structure in the known universe. The study of its structure and function and how it figures into our actions and mental experience is among the most exciting projects of modern science. This class begins with molecules and cells, builds up to brains and nervous systems, encompasses neural signaling, sensory perception, memory, language, and emotion, and culminates with the great mystery of how brain processes relate to consciousness and mental experience — that is, how mind may be related to brain. This is a comprehensive introduction to the exciting subject of contemporary neuroscience, open to all interested students.

Learning Goals

By the end of this course, you will be able to...

1. **Explain the building blocks of nervous system function**, from neurons and glia to action potentials, synaptic transmission, and neurochemical signaling.

2. **Trace how the brain generates cognition and behavior**, from sensory systems to memory, language, emotion, motivation, and social behavior.
3. **Understand how neuroscientists think and ask questions about the brain**: what makes a question scientifically tractable, how methods for probing the brain are matched to specific questions, and what different approaches can and cannot tell us.
4. **Engage with philosophical and psychological concepts of the mind**, considering how mental experience, consciousness, and behavior relate to the physical workings of the brain.
5. **Reason and argue about contested questions in neuroscience**, including how brain relates to mind, connecting course concepts to health, society, and everyday life through written reflections and in-class debates.

Materials

Required Textbook: Neuroscience: Exploring the Brain 5th Ed. Mark F. Bear, Barry W. Connors, Michael A. Paradiso. ISBN: 1284286878

This book is available at the campus bookstore, A free E-Book is also available online (CalCentral log-in required) for a maximum of 3 users at a time at [this](#) link.

Most course information is provided on the [course website](#). bCourses will be used for announcements, submitting assignments, grading, and posting lecture slides.

Course Calendar

Week	Date	Topic	Reading	Assignments
1	8/25	<i>No Class</i>	--	
	8/28	Intro/History/What is mind?	Chapter 1	
2	9/1	Neuroanatomy, Cells	Chapter 2, Chapter 7	
	9/4	Ion channels, membrane potential	Chapter 3	
3	9/8	Action potential	Chapter 4	
	9/11	Synaptic Transmission	Chapter 5	Pre-Course Survey Due
4	9/15	Neurotransmitters	Chapter 6	

	9/18	Drugs	Chapter 15	<i>Reflection 1 in lecture</i>
5	9/22	Development/Plasticity	Chapter 23	
	9/25	<i>Review Session or Guest Lecture</i>	--	
6	9/29	MIDTERM 1	--	
	10/2	Probing the Brain	p86 (electrodes), p90 (optogenetics), p203-209 (new views of the brain), p705-712 (EEG, EMG, ECoG), p840 (TMS), p843-844 (DBS),	
7	10/6	Vision 1	Chapter 9	<i>Debate 1 in Section</i>
	10/9	Vision 2	Chapter 10	
8	10/13	Olfaction	Chapter 8	
	10/16	Taste	Chapter 8	<i>Reflection 2 in lecture</i>
9	10/20	Hearing	Chapter 11	
	10/23	Somatosensation	Chapter 12	
10	10/27	<i>Review Session or Guest Lecture</i>	--	
	10/30	MIDTERM 2	--	
11	11/3	Language	Chapter 20	
	11/6	Memory Systems	Chapter 24	
12	11/10	Cellular Mech of Memory	Chapter 25	
	11/13	Sleep	Chapter 19	<i>Reflection 3 in lecture</i>
13	11/17	Emotions	Chapter 18	<i>Debate 2 in Section</i>
	11/20	Motivated Behavior	Chapter 16	
14	11/24	Social Behavior	<i>TBD</i>	
	11/27	<i>No Class</i>	--	
15	12/1	Mental Health	Chapter 22	
	12/4	Consciousness + Intelligence	Chapter 21	<i>Reflection 4 in lecture</i>
	12/17	FINAL EXAM, 11:30a	--	

Assignments & Course Components

- In-class Written Reflections: We will have four in-class written reflections designed to help you think more deeply about the applications and implications of course topics. In-class assignments can only be made up with an excused absence such as a documented medical reason.
- Exams: We will have three exams – two midterms in Weeks 6 and 10, and a final exam on 12/17. These will be multiple choice scantron exams. We will hold review sessions before each exam. Make-ups are not permitted for exams except in the case of a medical emergency.
- Debates: There will be two participatory debates that will take place during discussion section. Details of the debate format will be posted in Files: Debate. The debates will take place during Weeks 7 and 13. Topics will be decided as the semester progresses. We endeavor to develop debate topics that are relevant and timely, and for which strong arguments can be made for both sides. If you need to miss a debate for an excused reason, please contact your GSI to complete a make-up assignment.
- Discussion Section Participation: Discussion section attendance and participation are a required part of your grade, and you are strongly encouraged to attend. Your GSIs will provide more information about how to earn points for discussion section participation.
- Lecture Clicker Participation: I aim to make lecture a place of active learning and engagement. You will work with other students to discuss the material and work on practice problems. To encourage attendance and active participation we will be using the iClicker polling system and giving credit for using it during lectures.

You will receive .33 points for answering 50% (or more) of the questions in each class period. There are 24 possible lectures where you can earn clicker points, however we will set the maximum total points for lecture participation to 6 points (18 lectures). In other words, you can miss six lectures with no consequence to your lecture score. There will not be any make-up assignments for missing lecture since you basically get several free absences.

You are required to participate with the free iClicker student app on a smartphone, tablet or laptop (not with a physical remote). It is your responsibility to follow the steps to properly get added to our iClicker course in a timely fashion. Using your iClicker for participation points while you are not physically present in class is considered cheating and will result in loss of all your participation points for the semester (this is surprisingly easy to catch). It is also your responsibility to regularly check your iClicker records for any discrepancies and bring them to the attention of your GSI within 48 hours.

We will start using iClickers during the third lecture on September 4th, so make sure you have your account set up by then. Please follow the instructions on bCourses and the course website to get your account set up.

Grading

Category	Course Component	Points
Misc.	Pre-course Survey	1
	Course Evaluation (>80% class participation)	1
Assignments	In-class Reflections (4 x 4 pts each)	16
	Debates (2 x 7 pts each)	14
Exams	Midterms (2 x 15 pts each)	30
	Final	25
Participation	Section Participation	10
	Lecture Clicker Participation (18 x .33 pts)	6
	TOTAL:	103

Your final letter grade in this course will be based on a point accumulation system, rather than a percentage system. The total number of available points is 103, which means there are 3 buffer points built in. These exist to give you flexibility for an off week, an unexpected conflict, or simply an aspect of the course you want to engage with less than others. Nothing in this course is strictly mandatory; you have full agency over your learning and can choose how to invest your effort. I encourage you first and foremost to focus on expanding your thirst for knowledge, and enjoying the process of doing so, trusting that good grades will come.

Points Earned	Letter Grade
>97	A+
93-96	A
90-92	A-
87-89	B+
83-86	B
80-82	B-
77-79	C+
73-76	C
70-72	C-
60-69	D
<59	F

PLEASE NOTE: Because the 3 point buffer is built in, we won't be offering extra credit, rounding borderline grades, or accepting make-up work for unexcused absences. Please do not email course staff with such requests. That's what the 3 points are for.

Policies & FAQ

Attendance Policy

We aim to make lecture as useful as possible by doing interactive exercises and activities in addition to lecture – hopefully you find it worthwhile to attend. You have the opportunity to earn credit towards your grade by attending lecture, answering clicker questions, and completing four in-class reflections throughout the semester. Make-ups for these assignments will not be offered unless you have a documented medical absence. Section attendance and participation is highly encouraged and earns you 10 points toward your grade.

Late Assignments Policy

Assignments may be submitted up to 5 days late, with a 10% deduction per late day (maximum deduction: 50%).

What happens if I need to miss an exam?

If you know you need to miss an exam for a legitimate reason (illness, emergency, etc.), please contact your GSI ASAP. While we do not offer make-up exams of any kind, in this circumstance, your final exam will count as a proportionally larger percent of your grade. A documented excuse will be required, and this policy is only available for emergencies.

What should I do if I need learning accommodations?

Please make sure to have your accommodations noted with the Disabled Students Program. If your accommodations enable you to take exams under specific conditions, please make sure to contact the DSP office one week in advance of each exam to ensure you will be able to take your exam in the DSP room. Contacting DSP on time is your responsibility, and course staff do not have control over DSP testing space if the deadline is missed.

What is the academic integrity policy for this course?

Academic dishonesty is forbidden and is reportable to the Center for Student Conduct. If you are struggling with course material, please reach out to me or the GSI so we can help you. Resorting to cheating is unfair to other students, will create bad habits in yourself, and leave you unprepared to tackle future challenges in school and in work.

The following are considered academic dishonesty: plagiarism, copying answers from someone else during an exam, talking to another student during an exam, using your phone or other technology during closed-book assessments, faking your or another student's attendance, copying AI-generated material and submitting it as your own work, altering answers on a graded assessment and then asking for a re-grade.

Assignments in which academic dishonesty are discovered will be recorded as a zero.

How do I provide feedback about this course?

This is the first time I'm teaching NEU C61. I've put a lot of thought into how it's designed, including the AI guided tools, the structure of assignments, and the grading approach. I would love to know what's working for you and what isn't. You can share feedback with me directly in office hours or by email. If you'd prefer to do so anonymously, you can use [this](#) feedback form at any point in the semester.

Can I Use AI to Help Me in This Course?

Yes, you are welcome (and encouraged!) to use AI to help you learn in this course. I use AI regularly in my own work, and I've spent considerable time thinking about how to use it well in the classroom. The interactive tools on the course website are built to help you learn how to use AI as a learning partner.

That said, **there's an important difference between using AI to deepen your understanding and using it to avoid thinking altogether.** The first will make you a stronger learner. The second will leave you unprepared, not just for this course, but for the independent thinking that work and life require. AI tools are also imperfect: they hallucinate, oversimplify, and reflect the biases of their training data. Treating their output as truth without verifying is a habit you'll want to break sooner rather than later.

Throughout the course I'll share examples of how I use AI to clarify ideas, research sources, get a broad-strokes understanding of a research area, stress-test my thinking, and get feedback on my writing. I encourage you to experiment in the same spirit. **Submitting AI-generated text as your own work is not permitted under any circumstance. Not only is this very easy to spot, it's a waste of the learning opportunity you have in front of you.**